TNO innovation for life

HOW TO MAKE ENERGY FLEXIBILITY IN THE BUILT ENVIRONMENT WORK

APPLYING THE S2 (EN50491-12-2) STANDARD IN PRACTICE

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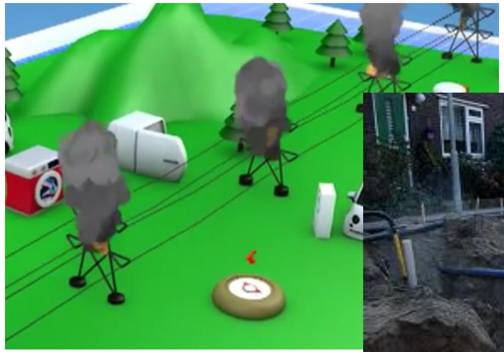
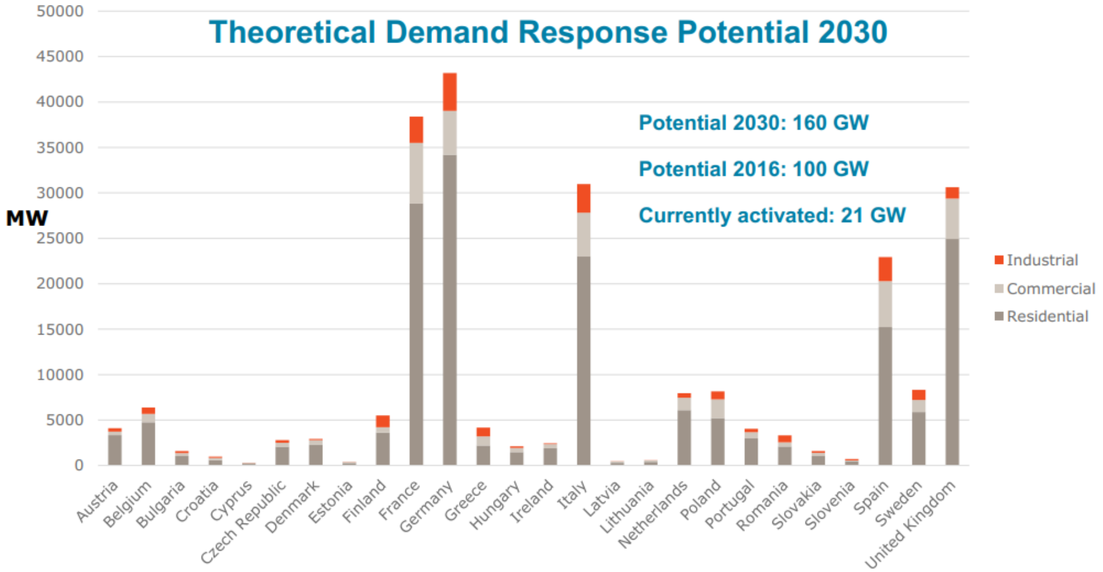
ENERGY FLEXIBILITY IN THE BUILT ENVIRONMENT

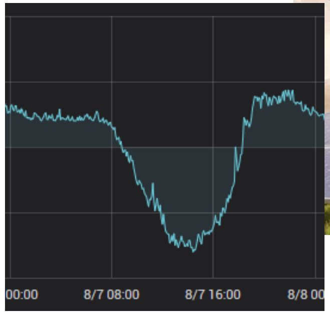


- Energy flexibility in the built environment has a great potential to help balance the power grid and prevent local congestion in the power grid
  - Congestion created by EV charging and heat pumps can be solved by flexible EV chargers and heat pumps
- In order for it to be successful on the long term, we need to make sure that:
  - User comfort has priority
  - Once its installed the user doesn't have to think about it: it needs to be automated

Flexible generation requires flexible demand to reduce peak electricity prices and system costs



Source: Impact Assessment support Study on downstream flexibility, demand response and smart metering, COWI, 2016



IN ORDER TO MAKE ENERGY FLEXIBILITY IN THE BUILT ENVIRONMENT SUCCESSFUL, CONNECTING DEVICES NEEDS TO BE...

CHEAP

RELIABLE

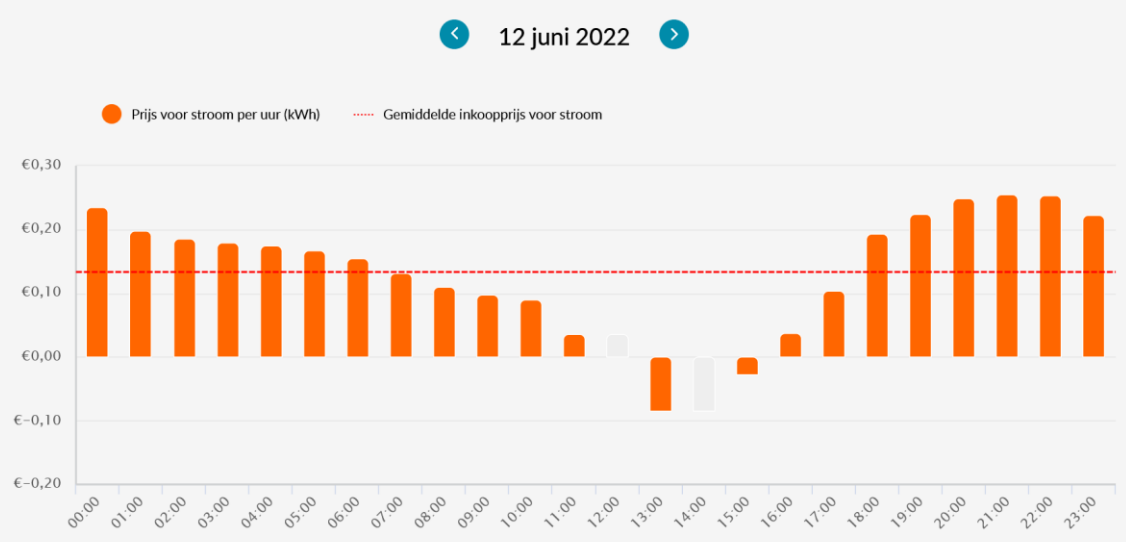
SECURE

DIVERSITY IN DEVICES

- In the built environment are many interesting flexible devices, such as…
  - EV chargers
  - Heating Ventilation & Air Conditioning (HVAC)
  - Home batteries
  - Curtailable PV systems

- But there are…
  - Different types of devices
  - Different brands
  - Different models with different API's

DIVERSITY IN OPTIMIZATIONS



- **Currently feasible optimizations**
  - Reduce costs by utilizing day and night tariffs
  - Capacity main fuse (e.g. stay below 3 x 25A)
  - Reduce costs by utilizing hourly tariffs
- **Future optimizations**
  - Self-consumption due to phasing out of net metering in the Netherlands (“salderen”)
  - Congestion management through DSO
  - Aggregator with energy/balancing/congestion market optimization (value stacking)
  - Real-time P2P energy trading
  - Whatever the future brings…

- Optimizations are tightly coupled with the (financial) incentive the consumer has to do something with flexibility!

NO ONE CAN DO IT ALONE

CURRENT SITUATION

Flexibility function from the manufacturer
Some manufacturers create flexible services with their devices. However, they typically only support one type of optimization, which doesn't make the products future proof.

- Home batteries that charge with locally produced energy until they are full
- EV chargers that limit charging power when the capacity of the grid connection is reached

Independent flexibility enablers
Some companies create an IoT platform which connect to different types of devices from different manufacturers. However, they struggle to implement support for many devices and do this in a cost-effective manner.

- Typically a gateway combined with a cloud infrastructure
- Measuring and providing insight usually the primary use case
- Typically for pilot projects, not yet commercially feasible

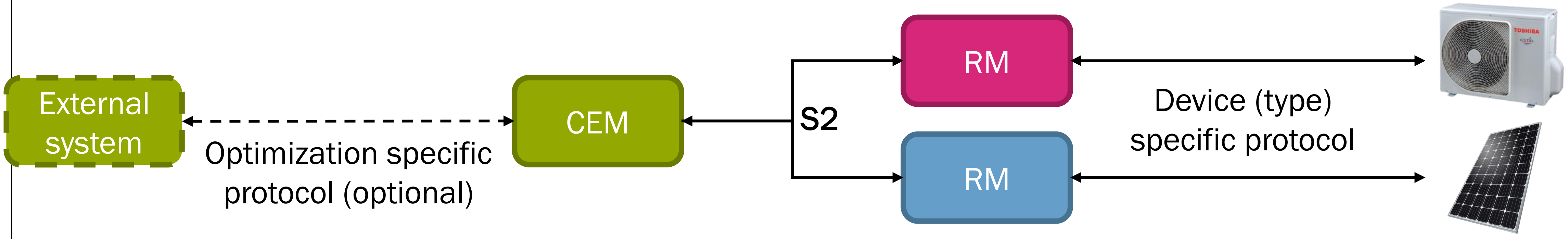
When can I buy any random device in a store and connect it to my EMS?

THE S2 (EN50491-12-2) APPROACH

- The S2 protocol has a unique approach
  - It doesn't assume what kind of device is being used
  - It doesn't assume what kind of optimization is used
  - Instead of describing how the device works or how the optimization works, S2 only describes the (potential) energy behaviour of the device
- The device stays in control
  - The device stays responsible for a safe operation and user comfort
  - The device decides which flexibility is given to the energy manager, and what is not
  - The device can always decide to ignore instructions from the energy manager

THE S2 (EN50491-12-2) APPROACH

- S2 splits the energy management system into two *apps*:
  - The Resource Manager (RM) handles the energy management for one device
  - The Customer Energy Manager (CEM) handles the optimization for the building
- A device which is controlled through S2 can easily be upgraded to a new type of optimization by connecting it to a different CEM, making the device future proof
  - The optimization is in the software, not in the hardware or firmware
- The CEM and RM can use other (existing) protocols to talk with other components
  - Existing connected devices can be retrofitted to S2 with a RM app



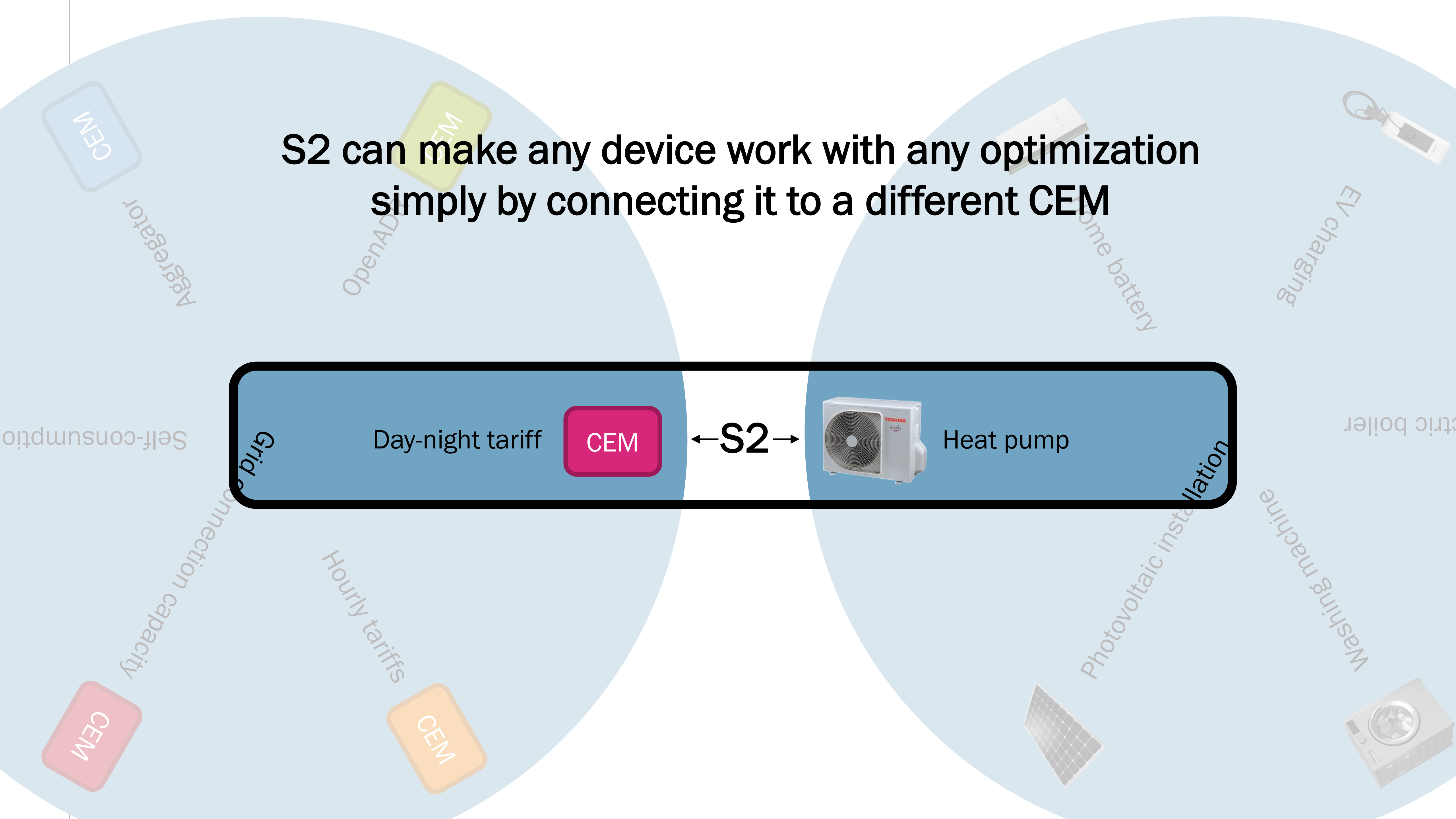
THE S2 (EN50491-12-2) APPROACH

- S2 can be used to send power measurements and power forecasts to the CEM
- A Resource Manager can offer the CEM one or multiple Control Types
  - A Control Type is a sub protocol for an abstract type of flexibility
  - A CEM typically implements all five, a device can implement one

CEM

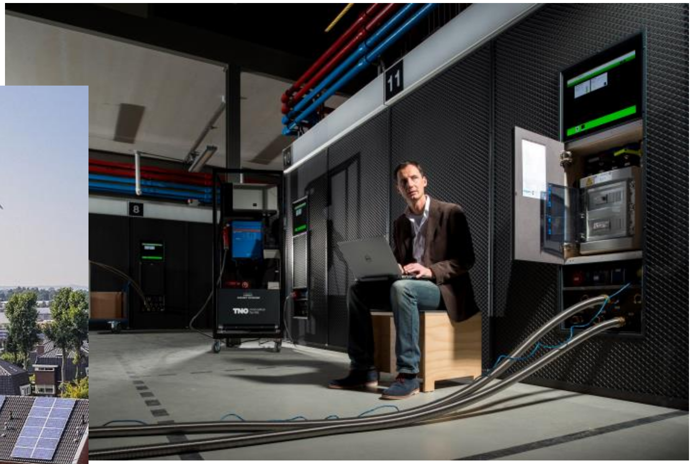
Control Type	Typical example
Power Envelope Based Control	PV curtailment, EV charging
Power Profile Based Control	Dishwasher, Dryer, Washing machine
Operation Mode Based Control	Diesel generator
Fill Rate Based Control	EV charging, Home battery, All electric heat pump
Demand Driven Based Control	Hybrid heat pump

S2 can make any device work with any optimization simply by connecting it to a different CEM



S2 ORIGIN: EFI

- S2 is based on the Energy Flexibility Interface (EFI), which is developed by TNO
- As an independent research organisation, TNO can oversee the whole system
- We foresaw the need for interoperability early on
- EFI is the result of ten years research and experience with energy flexibility
  - It was redesigned many times after testing it in practice
- Has been proven in many pilot projects and lab settings together with many project partners and many types of devices and optimizations



THE GO-E PROJECT

- No one can do it alone!
- That's why in the Dutch GO-e project DSO's, energy companies, IT companies and research institutes work on:
  - Defining scalable flexibility services that allow better utilization of the existing power grid
  - Unlocking flexibility in an cheap, reliable and secure way
  - Insights into (future) power grid congestion so DSO's can make informed decisions
  - Consumers and businesses are at the centre of in the design of flexibility services

Klankbord

Living Labs: Zakelijk:

Living Labs: Consumenten:

- For more information see www.projectgo-e.nl (Dutch)

GO-E OBJECTIVES FOR UNLOCKING FLEXIBILITY

- The standardization of S2 is not enough to ensure a cheap, reliable and secure way of unlocking flexibility
- The unlocking flexibility work package wants to help by
  - **Creating an architecture around S2**
    S2 is only a part of what's needed for an energy management system. GO-e develops an IT architecture for energy management systems with a wider scope (for those who want it). We don't want to reinvent the wheel, but instead provide an architecture that can be integrated into existing solutions (such as home automation systems).
  - **Providing practical way of working with S2**
    S2 is a semantical standard, which can result in multiple (mutually compatible) protocol stacks. GO-e is developing a well defined S2 version suitable for use in LAN and Internet. Also developer resources such as documentation and debugging tooling will be developed.
  - **Coming up with an adoption plan**
    Look at business interest of all stakeholders and see which conditions will have to be met for a widespread adoption. GO-e is also looking for partners to collaborate with.

- We're still looking for members for our advisory committee! Let us know if you want to get involved!

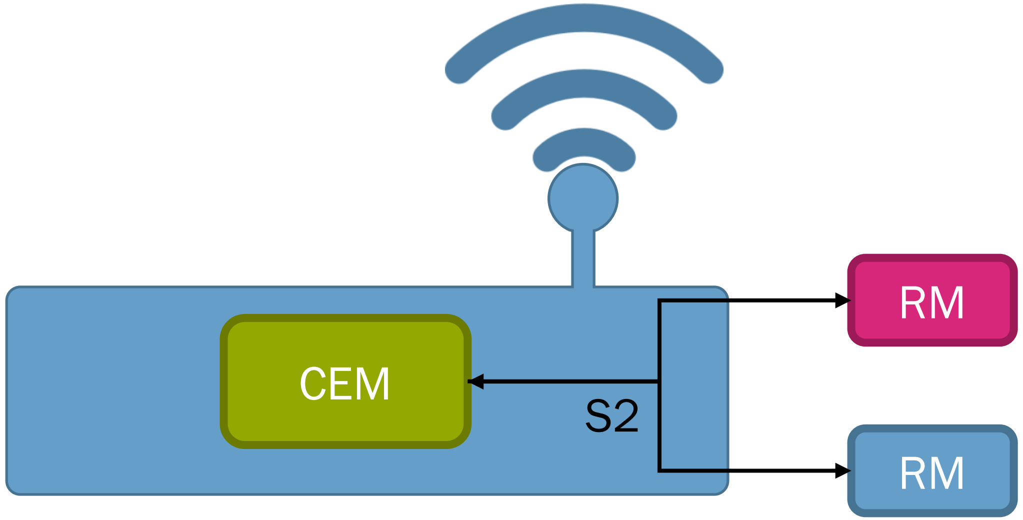
RELATION WITH IOT

- Energy management is essentially an IoT application
- The main motivation manufacturers to make their devices connected is often not flexibility, but monitoring or home automation
  - Smart devices sell because their cool features, not (yet) because of their energy flexibility features
  - This provides on opportunity to piggy-bag energy flexibility on the connectedness of devices!
- There are several home automation platforms out their that already connect to smart devices
  - There is an opportunity utilize these platforms for Home Energy Management System (HEMS) functionality

LOCATION OF THE CEM

Inside a HEMS device

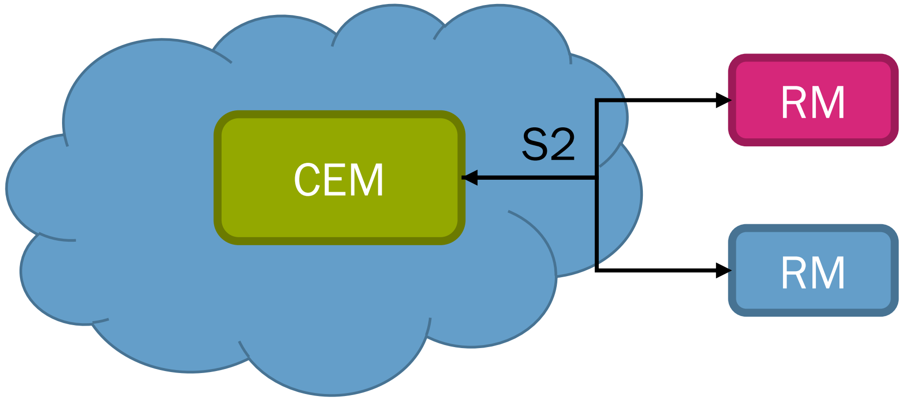
The CEM is an app installed in a small computer in the building



+ Doesn't require a (sometimes unreliable) internet connection
+ Suitable for use cases where quick response and reliability is required (e.g. capacity grid connection)

Inside the cloud

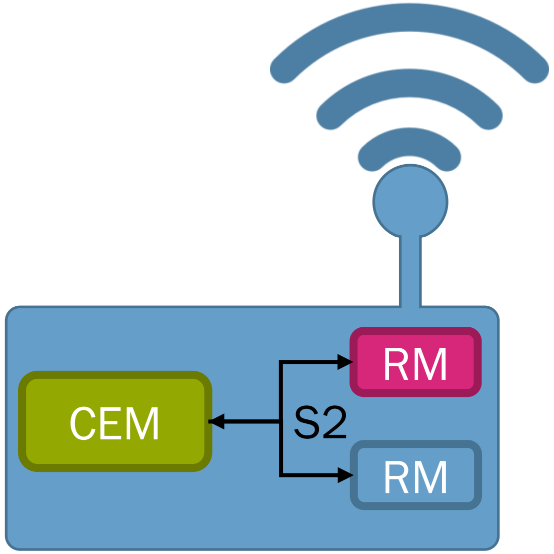
The CEM is an online service



+ No hardware costs for the consumer
+ Easier to maintain by developer
+ Easier to troubleshoot

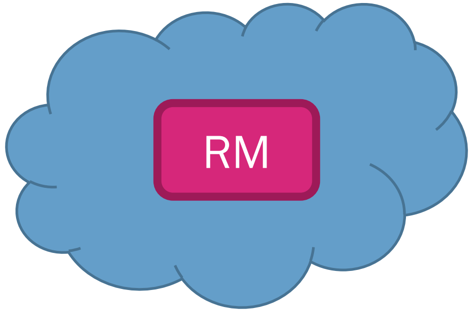
LOCATION OF THE RESOURCE MANAGER

Inside a HEMS device



S2 communication is internal, RM communicates with device through device (type) specific protocol

Inside the manufacturer cloud

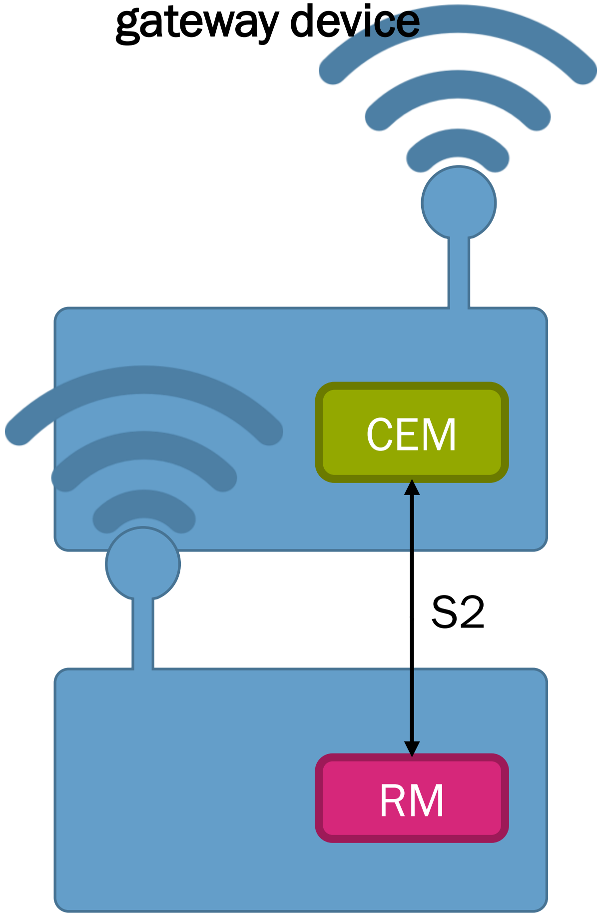


Many smart devices communicate directly through WiFi, GPRS, LoRa etc. to the cloud back-end of the manufacturer

Inside the device

Computation power on devices is inexpensive nowadays, the manufacturer can make it part of the device

Inside a separate gateway device



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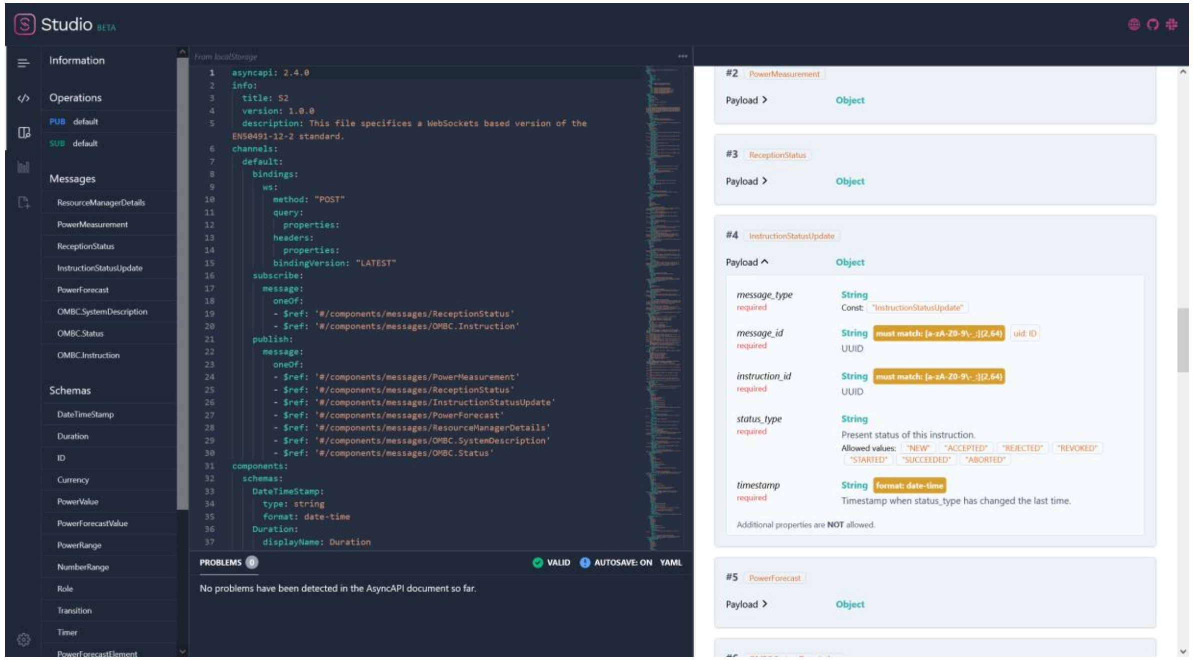
AN IP BASED VERSION OF S2



- Requirements for an IP based version of S2
  - Should support all deployment scenarios (keeping NAT and firewalls in mind)
  - Use technologies familiar to most developers
  - Event driven (no polling necessary)
  - Mature
  - Scalable
  - Secure
- After thorough consideration we decided to go with JSON messages over WebSockets
  - Communication described in the AsyncAPI standard (www.asyncapi.com)
  - AsyncAPI is similar to OpenAPI (Swagger), but works with messages
  - Messages are described in JSON-schema
  - Generators for many programming languages available

DEVELOPER TOOLING

- Documentation explaining the S2 concepts
- Examples for typical flexible devices
- Tooling to test your implementation
- Protocol analyser for debugging, visualizing the communication and detecting common problems

- For updates check www.projectgo-e.nl and sign up for the FAN newsletter (www.flexible-energy.eu)!

CONCLUSION

- Energy flexibility is the built environment has yet to take of because
  - There are many different types, brands and models of devices that can provide flexibility
  - There are many options for optimizing the flexibility and providing an incentive to the consumer
- Energy flexibility in the built environment is only going to work if it’s cheap and easy to unlock, and flexible when it comes to devices and optimizations
- The S2 protocol has the potential to solve the interoperability problem, but only if its adopted by devices and energy management systems
  - The GO-e project aims to promote the S2 standard by providing the right technical tools and creating a network of companies that want to collaborate

  - **We are still looking for members for the advisory committee. If you want to join us on this mission, let us know!**

THANK YOU FOR YOUR TIME

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